

Calc 3 – Partial Derivatives

Worksheet A

1. For $f(x, y) = x^3y^2 + \sin(xy)$, find f_x and f_y .
2. Compute f_{xy} and f_{yx} for $f(x, y) = e^{xy} + x^2y$ and verify Clairaut's theorem.
3. Show that $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2y}{x^4 + y^2}$ does not exist (try $y = x^2$).
4. Find the tangent plane to $z = x^2 + y^2$ at $(1, 2, 5)$.
5. Use the chain rule: $z = x^2 + y^2$, $x = \cos t$, $y = \sin t$. Compute dz/dt .

Calc 3 – Partial Derivatives

Worksheet B

1. For $f(x, y, z) = xyz + z^2e^x$, compute all three first partials.
2. For $z = \ln(x^2 + y^2)$, verify Laplace's equation $z_{xx} + z_{yy} = 0$.
3. Determine whether $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 + y^2}{|x| + |y|}$ exists.
4. Find the tangent plane to the surface $x^2 + y^2 + z^2 = 14$ at $(1, 2, 3)$.
5. Use implicit differentiation on $F(x, y, z) = x^2 + 2y^2 + 3z^2 - 6 = 0$ to find $\partial z / \partial x$ at $(1, 1, 1)$.

Calc 3 – Partial Derivatives

Answer Key (Worksheets A & B)

Worksheet A.

1. $f_x = \boxed{3x^2y^2 + y \cos(xy)}$, $f_y = \boxed{2x^3y + x \cos(xy)}$.
2. $f_x = ye^{xy} + 2xy$, $f_{xy} = e^{xy} + xye^{xy} + 2x$. $f_y = xe^{xy} + x^2$, $f_{yx} = e^{xy} + xye^{xy} + 2x$. Equal. ✓
3. Along $y = mx$: $\frac{x^2 \cdot mx}{x^4 + m^2x^2} = \frac{mx^3}{x^4 + m^2x^2} \rightarrow 0$. Along $y = x^2$: $\frac{x^2 \cdot x^2}{x^4 + x^4} = \frac{1}{2}$. Different limits $\Rightarrow$ does not exist.
4. $f_x = 2x$, $f_y = 2y$. At $(1, 2)$: gradient $(2, 4)$. Plane: $z - 5 = 2(x - 1) + 4(y - 2) \Rightarrow \boxed{z = 2x + 4y - 5}$.
5. $\frac{dz}{dt} = 2x \cdot (-\sin t) + 2y \cdot \cos t = -2 \sin t \cos t + 2 \sin t \cos t = \boxed{0}$ (consistent with $z = 1$ constant).

Worksheet B.

1. $f_x = \boxed{yz + z^2e^x}$, $f_y = \boxed{xz}$, $f_z = \boxed{xy + 2ze^x}$.
2. $z_x = \frac{2x}{x^2+y^2}$, $z_{xx} = \frac{2(x^2+y^2)-2x \cdot 2x}{(x^2+y^2)^2} = \frac{2(y^2-x^2)}{(x^2+y^2)^2}$. By symmetry $z_{yy} = \frac{2(x^2-y^2)}{(x^2+y^2)^2}$. Sum = 0. ✓
3. Along $y = 0$: $\frac{x^2}{|x|} = |x| \rightarrow 0$. Along $y = x$: $\frac{2x^2}{2|x|} = |x| \rightarrow 0$. In fact $\frac{x^2+y^2}{|x|+|y|} \leq |x| + |y| \rightarrow 0$ (since $x^2 + y^2 \leq (|x| + |y|)^2$). **Limit** = 0.
4. $\nabla F = \langle 2x, 2y, 2z \rangle$. At $(1, 2, 3)$: $\langle 2, 4, 6 \rangle$. Plane: $\boxed{2(x - 1) + 4(y - 2) + 6(z - 3) = 0}$ i.e. $x + 2y + 3z = 14$.
5. $\frac{\partial z}{\partial x} = -\frac{F_x}{F_z} = -\frac{2x}{6z}$. At $(1, 1, 1)$: $\boxed{-1/3}$.